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# =====
# Quantization in Grayscale Images
# Google Colab Compatible
# =====

# Install OpenCV
!pip install opencv-python -q

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# -----
# Upload Image
# -----
print("Upload a grayscale or color image")

uploaded = files.upload()

filename = list(uploaded.keys())[0]

# Read image in grayscale
img = cv2.imread(filename, cv2.IMREAD_GRAYSCALE)

if img is None:
    print("Error: Image not loaded.")
else:

    # -----
    # Quantization Function
    # -----
    def quantize(image, levels):
        step = 256 // levels
        quantized = (image // step) * step
        return quantized.astype(np.uint8)

    # Different Quantization Levels
    img_256 = img # Original (256 levels)
    img_64 = quantize(img, 64)
    img_16 = quantize(img, 16)
    img_8 = quantize(img, 8)
    img_4 = quantize(img, 4)

    # -----
    # Display Results
    # -----
    plt.figure(figsize=(15,8))

    plt.subplot(2,3,1)
    plt.imshow(img_256, cmap='gray')
    plt.title("256 Gray Levels")
    plt.axis("off")

    plt.subplot(2,3,2)
    plt.imshow(img_64, cmap='gray')
    plt.title("64 Gray Levels")
    plt.axis("off")

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plt.subplot(2,3,3)
plt.imshow(img_16, cmap='gray')
plt.title("16 Gray Levels")
plt.axis("off")

plt.subplot(2,3,4)
plt.imshow(img_8, cmap='gray')
plt.title("8 Gray Levels")
plt.axis("off")

plt.subplot(2,3,5)
plt.imshow(img_4, cmap='gray')
plt.title("4 Gray Levels")
plt.axis("off")

plt.tight_layout()
plt.show()

# -----
# Conclusion
# -----
print("\nObservation:")
print("• 256 Levels → Highest image quality and detail.")
print("• 64 Levels → Slight loss of detail.")
print("• 16 Levels → Noticeable banding begins.")
print("• 8 Levels → Significant detail loss.")
print("• 4 Levels → Severe banding and poor image quality.")

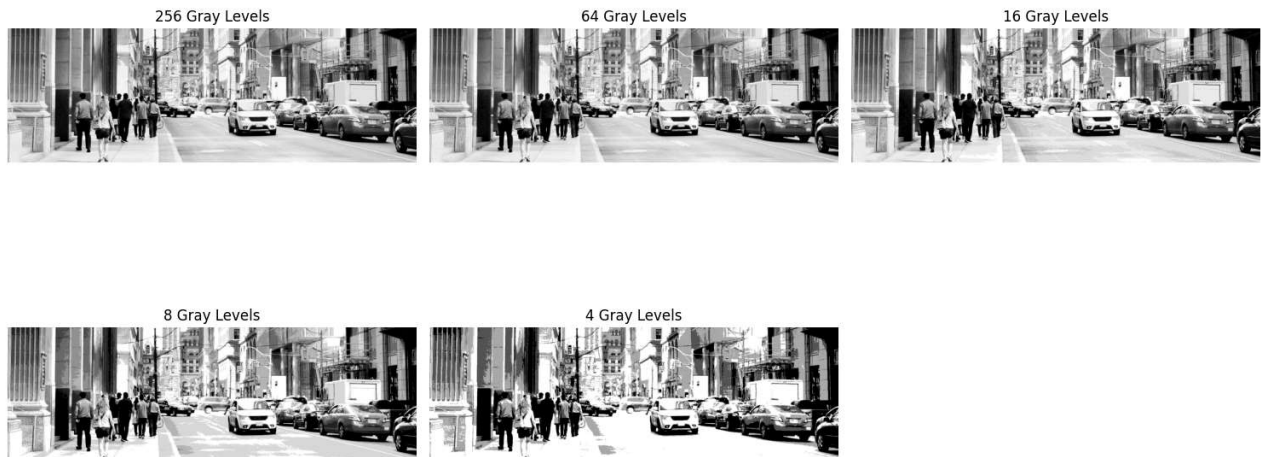
```

Upload a grayscale or color image

number_plate.jpg

number_plate.jpg(image/jpeg) - 123440 bytes, last modified: 7/13/2026 - 100% done

Saving number_plate.jpg to number_plate.jpg



Observation:

- 256 Levels → Highest image quality and detail.
- 64 Levels → Slight loss of detail.
- 16 Levels → Noticeable banding begins.
- 8 Levels → Significant detail loss.
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